Mathematical Foundations of Autonomic Delay-Tolerant Network Management Architecture (DTNMA) Routing

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Abstract—Delay- and Disruption-Tolerant Networks (DTNs) deployed in deep space and terrestrial tactical domains face severe intermittent connectivity, prolonged orbital round-trip times, and extreme buffer contention. Centralized network management protocols (e.g., SNMP, NETCONF) fail catastrophically under these conditions. This paper presents the comprehensive mathematical formulation of the autonomic DTNMA routing framework conforming to RFC 9675 (DTNMA) and RFC 7575 (Autonomic Networking). We derive the state-space dynamics of strongly-typed Externally Defined Data (EDD) telemetry, model probabilistic encounter recurrences, establish formal proofs for binary token conservation and single-copy invariants, and formulate constructive ingress admission control and backpressure queue gradient gating. Furthermore, we provide a formal proof of the Inviolable Zero-Drop Theorem for high-priority fault telemetry and prove the asymptotic Lyapunov stability of the closed-loop autonomous system under severe buffer congestion.

Index Terms—Delay-Tolerant Networking (DTN), DTNMA, RFC 9675, Autonomic Networking, Binary Token Spraying, Backpressure Control, Zero-Drop Invariant, Lyapunov Stability.

1 Introduction and System Model

Conventional terrestrial networking paradigms rely upon persistent end-to-end paths and bounded round-trip latencies. In contrast, deep-space, vehicular (VDTN), and aerial (FANET) networks operate under time-varying, episodic topology graphs where instantaneous paths between sources and destinations rarely exist. To provide automated, survivable operations in the absence of real-time human control, the Delay-Tolerant Network Management Architecture (DTNMA, RFC 9675 [1]) introduces asynchronous, declarative management via State-Based Rules (SRLs).

1.1 Network Graph Formalism

We model the network as an episodic, time-varying contact graph:

$$\mathcal{G}(t) = \big(\mathcal{V}, \mathcal{E}(t)\big) \tag{1}$$

where $\mathcal{V} = \{1, 2, \ldots, N\}$ denotes the set of N autonomous nodes, and an undirected communication edge $(i, j) \in \mathcal{E}(t)$ exists at time t if and only if:

$$\|\mathbf{p}_i(t) - \mathbf{p}_j(t)\|_2 \leq R_{\text{tx}} \tag{2}$$

Internal R&D Technical Specification. Corresponding module implementations in NASA/SCaN DTNMA Open-Source Framework.

where $\mathbf{p}_i(t) = [x_i(t), y_i(t)]^T \in \mathbb{R}^2$ represents the spatial coordinates of node i, and R_{tx} is the effective radio transmission range.

1.2 Node and Message State Spaces

Each autonomous node $i \in \mathcal{V}$ is defined by the state tuple:

$$S_i(t) = \big\langle S_{\max,i}, \mathcal{B}_i(t), \mathbf{x}_i(t), \mathcal{R}_i, \mathbf{u}_i(t) \big\rangle \tag{3}$$

where $S_{\max,i} \in \mathbb{R}^+$ is the total buffer memory capacity in bytes, $\mathcal{B}_i(t)$ is the current message storage set, $\mathbf{x}_i(t)$ is the continuous telemetry feature vector, $\mathcal{R}_i$ is the ordered rule database, and $\mathbf{u}_i(t)$ is the active policy actuation vector.

The message buffer $\mathcal{B}_i(t) = \{m_1, m_2, \ldots, m_{K_i(t)}\}$ stores bundles, where each message m is characterized by:

$$m = \big\langle \text{id}(m), s(m), d(m), \ell(m), \tau_{\text{gen}}(m), \\ \text{ttl}(m), h(m), C(m), \text{class}(m) \big\rangle \tag{4}$$

Here, $s(m)$ is the originating source, $d(m)$ is the destination, $\ell(m)$ is the payload length in bytes, $\tau_{\text{gen}}(m)$ is the generation epoch, $\text{ttl}(m)$ is the remaining time-to-live, $h(m) \in \mathbb{N}_0$ is the accumulated hop count, and $C(m) \in \mathbb{N}^+$ denotes the remaining spray token quota. The traffic class is partitioned into two distinct priority sets:

$$\text{class}(m) \in \{\mathcal{C}_0, \mathcal{C}_1\} \tag{5}$$

where $\mathcal{C}_0$ denotes *Class 0: Critical Fault Observability Telemetry*, and $\mathcal{C}_1$ represents *Class 1: Bulk Science Payload*.

2 Strongly-Typed EDD Telemetry Vector Dynamics

Conforming to RFC 9675, node i periodically computes an 8-dimensional Externally Defined Data (EDD) telemetry vector:

$$\mathbf{x}_i(t) = \big[Q_i(t),\ \dot{Q}_i(t),\ v_i(t),\ D_i,\ E_i(t),\ \dot{E}_i(t),\ k_i(t),\ Q^{\mathcal{N}}_{\min,i}\big]^T \tag{6}$$

sampled at discrete time steps with interval $\Delta t \geq 0.5\,\text{s}$.

2.1 Normalized Buffer Occupancy Ratio ($Q_i(t)$)

Let $S_{\text{free},i}(t)$ be the currently unallocated buffer memory. The dimensionless occupancy ratio $Q_i(t) \in [0.0, 1.0]$ is:

$$Q_i(t) = \frac{S_{\max,i} - S_{\text{free},i}(t)}{S_{\max,i}} = \frac{1}{S_{\max,i}} \sum_{m \in \mathcal{B}_i(t)} \ell(m) \tag{7}$$

2.2 First-Derivative Queue Drift ($\dot{Q}_i(t)$)

To detect impending buffer collapse prior to packet loss, queue drift $\dot{Q}_i(t) \in \mathbb{R}\ [\mathrm{s}^{-1}]$ is computed via backward finite difference:

$$\dot{Q}_i(t) = \frac{Q_i(t) - Q_i(t - \Delta t)}{\Delta t} \tag{8}$$

A drift $\dot{Q}_i(t) > 0$ indicates net buffer accumulation ($\lambda_{\text{in}} > \mu_{\text{out}}$), whereas $\dot{Q}_i(t) < 0$ signifies queue draining.

2.3 Universal Kinematic Velocity ($v_i(t)$)

Speed is determined universally across arbitrary coordinate movement models:

$$v_i(t) = \frac{\|\mathbf{p}_i(t) - \mathbf{p}_i(t - \Delta t)\|_2}{\Delta t} = \frac{\sqrt{\Delta x_i^2(t) + \Delta y_i^2(t)}}{\Delta t} \tag{9}$$

2.4 Sliding-Window Temporal Contact Density ($D_i(t, W)$)

Let $\tau_{i,j}^{\text{last}}$ denote the timestamp of the most recent physical contact between node i and peer j. The historical contact set over a backward sliding window $W = 300\,\mathrm{s}$ is:

$$\mathcal{H}_i(t, W) = \left\{ j \in \mathcal{V} \setminus \{i\} \;\middle|\; (t - \tau_{i,j}^{\text{last}}) \le W \right\} \tag{10}$$

Let $\mathcal{A}_i(t) = \{ j \in \mathcal{V} \mid (i,j) \in \mathcal{E}(t) \}$ represent currently active links. The composite temporal density $D_i(t) \in \mathbb{N}_0$ is defined as:

$$D_i(t) = \left| \mathcal{H}_i(t, W) \cup \mathcal{A}_i(t) \right| \tag{11}$$

2.5 Energy State-of-Charge and Drain Rate

Let $E_i(t) \in [0.0, 1.0]$ denote the normalized remaining battery energy. The non-negative energy drain rate $\dot{E}_i(t)$ is:

$$\dot{E}_i(t) = \max\left(0.0,\ \frac{E_i(t - \Delta t) - E_i(t)}{\Delta t}\right) \tag{12}$$

2.6 Neighborhood Minimal Buffer Occupancy ($Q_{\min,i}^{\mathcal{N}}(t)$)

For multi-hop backpressure decisions:

$$Q_{\min,i}^{\mathcal{N}}(t) = \begin{cases} \min_{j \in \mathcal{A}_i(t)} Q_j(t), & \text{if } \mathcal{A}_i(t) \neq \emptyset \\ 1.0, & \text{if } \mathcal{A}_i(t) = \emptyset \end{cases} \tag{13}$$

3 Encounter and Delivery Predictability Dynamics

The telemetry sampler maintains a PRoPHET (RFC 6693 [3]) predictability matrix $\mathbf{P} \in [0,1]^{N \times N}$.

3.1 Direct Encounter Update

Upon establishment of an active contact link $(i,j) \in \mathcal{E}(t)$:

$$P_{(i,j)}(t^+) = P_{(i,j)}(t^-) + \left(1 - P_{(i,j)}(t^-)\right) \cdot P_{\text{init}} \tag{14}$$

where initialization parameter $P_{\text{init}} = 0.75$.

3.2 Transitive Predictability Diffusion

During link discovery between i and j, transitive probabilities toward all mutual destinations $k \in \mathcal{V} \setminus \{i,j\}$ are propagated:

$$P_{(i,k)}(t^+) = P_{(i,k)}(t^-) + \left(1 - P_{(i,k)}(t^-)\right) \cdot P_{(i,j)}(t^+) \cdot P_{(j,k)} \cdot \beta \tag{15}$$

where transitivity constant $\beta = 0.25$.

3.3 Continuous Predictability Aging

Over periods of link absence, predictability decays geometrically:

$$P_{(i,j)}(t) = P_{(i,j)}(t_{\text{prev}}) \cdot \gamma^{\left(\frac{t - t_{\text{prev}}}{\Delta t_{\text{unit}}}\right)} \tag{16}$$

where aging discount factor $\gamma = 0.98$ and time unit quantum $\Delta t_{\text{unit}} = 30.0\,\mathrm{s}$.

4 Declarative Predicate Autonomy Engine and Arbitration

4.1 Predicate Logic Model

An RFC 9675 State-Based Rule $R_k \in \mathcal{R}_i$ is defined as a tuple:

$$R_k = \left\langle \mathrm{ARI}_k, \pi(R_k), \Phi_k, \mathcal{A}(R_k) \right\rangle \tag{17}$$

where ARI_k is the Autonomous Resource Identifier string, $\pi(R_k) \in \mathbb{Z}^+$ is the static priority weight, Φ_k is the Boolean predicate, and $\mathcal{A}(R_k)$ is the set of executable controls.

An elementary clause $\phi_{k,m}$ tests an EDD entry against a threshold scalar θ_m:

$$\phi_{k,m}\left(\mathbf{x}_i(t)\right) = \left(\mathbf{x}_i[\mathrm{EDD}_m] \;\bowtie\; \theta_m\right), \quad \bowtie \in \{>, <, \geq, \leq, ==, \neq\} \tag{18}$$

Composite predicates are formed via conjunction or disjunction:

$$\Phi_k^{\text{AND}} = \bigwedge_{m=1}^{M_k} \phi_{k,m}, \qquad \Phi_k^{\text{OR}} = \bigvee_{m=1}^{M_k} \phi_{k,m} \tag{19}$$

4.2 Priority Arbitration Mechanics

At each evaluation cycle ($f_{\text{eval}} = 1.0\,\mathrm{Hz}$), the active rule R_{k^*} is determined by deterministic priority maximization over all satisfied rules:

$$k^* = \arg \max_{k \in \{1, \ldots, |\mathcal{R}_i|\}} \left\{ \pi(R_k) \;\middle|\; \Phi_k\left(\mathbf{x}_i(t)\right) = \mathrm{TRUE} \right\} \tag{20}$$

The policy actuator updates the node control vector:

$$\mathbf{u}_i(t) = \begin{bmatrix} L_{\text{spray}}(t) & M_{\text{ingress}}(t) & T_{\text{telem}}(t) & B_{\text{backpressure}}(t) \end{bmatrix}^T \tag{21}$$

governed by the canonical three-tier regime map:

$$\mathbf{u}_i(t) = \begin{cases} \begin{bmatrix} 2 & 1 & 600.0 & 1 \end{bmatrix}^T, & \text{if } Q_i \geq 0.92 \ \vee \ E_i < 0.20 \\ \begin{bmatrix} 3 & 0 & 300.0 & 1 \end{bmatrix}^T, & \text{if } D_i \geq 4.0 \\ \begin{bmatrix} 6 & 0 & 60.0 & 0 \end{bmatrix}^T, & \text{if } Q_i < 0.85 \ \wedge \ D_i < 4.0 \end{cases} \tag{22}$$

5 Two-Phase Forwarding Dynamics

5.1 Phase 1: Dynamic Spray and Binary Halving

When a new bundle m is generated at source $s(m) = i$:

$$C_0(m) = \begin{cases} 8, & \text{if } m \in \mathcal{C}_0 \text{ (Critical Faults)} \\ L_{\text{spray}}(t) \in \{2, 3, 6\}, & \text{if } m \in \mathcal{C}_1 \text{ (Bulk Science)} \end{cases} \tag{23}$$

When node i transfers bundle m ($C(m) > 1$) to neighbor $j \neq d(m)$, tokens are split via the asymmetric ceiling/floor operator:

$$C_i(m) \leftarrow \max\left(1, \left\lceil \frac{C(m)}{2} \right\rceil\right) \quad \text{(Retained by sender)} \tag{24}$$

$$C_j(m) \leftarrow \max\left(1, \left\lfloor \frac{C(m)}{2} \right\rfloor\right) \quad \text{(Assigned to receiver)} \tag{25}$$

Theorem 1 (Token Conservation Invariance)**.** *For any spray transmission with initial quota $C(m) \geq 2$, the sum of tokens across the link is strictly invariant under the binary splitting operator:*

$$C_i(m) + C_j(m) = C(m) \tag{26}$$

Proof. Let $C \in \mathbb{N}_{\geq 2}$. By Euclidean division, C is either even ($C = 2k$) or odd ($C = 2k + 1$) for $k \in \mathbb{N}^+$.

1) If $C = 2k$:
$$\left\lceil \frac{2k}{2} \right\rceil + \left\lfloor \frac{2k}{2} \right\rfloor = k + k = 2k = C$$
2) If $C = 2k + 1$:
$$\left\lceil \frac{2k+1}{2} \right\rceil + \left\lfloor \frac{2k+1}{2} \right\rfloor = (k + 1) + k = 2k + 1 = C$$

Since $k \geq 1$, $\lfloor C/2 \rfloor \geq 1$, thus $\max(1, \cdot)$ remains inactive. Hence, $C_i(m) + C_j(m) \equiv C(m)$. □

Corollary 1 (Replication Depth Bound)**.** *The replication tree of bundle m forms a binary tree whose maximum depth $d_{\max}(m)$ is bounded by:*

$$d_{\max}(m) = \lceil \log_2 C_0(m) \rceil \tag{27}$$

5.2 Phase 2: Cognitive Focus Handoff ($C(m) = 1$)

When token halving terminates at $C(m) = 1$, replication ceases. Transfer to a non-destination peer j is permitted if and only if the joint focus predicate $\Psi_{\text{focus}}(m, i, j)$ evaluates to TRUE:

$$\begin{aligned} \Psi_{\text{focus}}(m, i, j) = \mathbb{I}\big(h(m) < h_{\max}\big) \wedge \mathbb{I}\big(P_{(j,d)} \geq P_{\text{base}}\big) \\ \wedge \mathbb{I}\big(P_{(j,d)} > P_{(i,d)} + \Delta P\big) \wedge \mathbb{I}\big(Q_j < 0.90\big) \end{aligned} \tag{28}$$

with parameters:

$$\langle h_{\max}, P_{\text{base}}, \Delta P \rangle = \begin{cases} \langle 3,\ 0.25,\ 0.10 \rangle, & \text{if } m \in \mathcal{C}_0 \\ \langle 2,\ 0.35,\ 0.15 \rangle, & \text{if } m \in \mathcal{C}_1 \end{cases} \tag{29}$$

Theorem 2 (Single-Copy Global Conservation Invariant)**.** *Under the cognitive focus handoff phase, the global network copy count satisfies:*

$$\sum_{k \in \mathcal{V}} \mathbb{I}\big(m \in \mathcal{B}_k(t)\big) \equiv 1, \quad \forall t \geq t_{\text{focus}} \tag{30}$$

Proof. Upon successful completion of a focus transfer from node i to node j, node i immediately executes:

$$\mathcal{B}_i(t^+) = \mathcal{B}_i(t^-) \setminus \{m\}$$

Concurrently, the receiving router forces:

$$C_j(m) \leftarrow 1$$

Because the local copy at sender i is deleted instantaneously upon link acknowledgment and receiver j cannot re-seed extra tokens, the indicator sum across all node buffers $\mathcal{V}$ remains exactly 1. □

6 Backpressure Gradient and Ingress Flow Control

6.1 Backpressure Queue Gradient Gating

Let $\Delta Q_{ij}(t) = Q_j(t) - Q_i(t)$ denote the spatial buffer occupancy gradient between nodes i and j. The forwarding gating function $G(m, i, j) \in \{0, 1\}$ is formulated as:

$$G(m, i, j) = \begin{cases} 1, & \text{if } j = d(m) \text{ (Direct delivery)} \\ 1, & \text{if } m \in \mathcal{C}_0 \text{ (Critical faults)} \\ 0, & \text{if } m \in \mathcal{C}_1 \ \wedge\ Q_j(t) \geq 0.95 \\ 0, & \text{if } m \in \mathcal{C}_1 \ \wedge\ B_{\text{bp}} = 1 \ \wedge\ \Delta Q_{ij} \geq 0.15 \\ 1, & \text{otherwise} \end{cases} \tag{31}$$

6.2 Constructive Ingress Admission Control

To prevent wasted channel transmission energy and link airtime from post-hoc dropping, incoming transfers are gated at the link handshake level:

$$\Gamma(m, i) = \begin{cases} \text{RCV_OK}, & \text{if } m \in \mathcal{C}_0 \\ \text{DENIED}, & \text{if } m \in \mathcal{C}_1 \ \wedge\ M_{\text{ing}} = 2 \ \wedge\ Q_i > 0.95 \\ \text{RCV_OK}, & \text{otherwise} \end{cases} \tag{32}$$

7 Dual-Tier Preemption and Inviolable Storage Eviction

7.1 Strict Total Priority Ordering

Bundles pending transmission are scheduled under a strict dual-tier total order $\prec$:

$$m_1 \prec m_2 \iff \begin{cases} \text{class}(m_1) = \mathcal{C}_0 \ \wedge\ \text{class}(m_2) = \mathcal{C}_1 \\ \text{class}(m_1) = \text{class}(m_2) \ \wedge\ \text{ttl}(m_1) < \text{ttl}(m_2) \end{cases} \tag{33}$$

This ensures Head-of-Line (HoL) preemption for all critical fault telemetry.

7.2 Inviolable Buffer Replacement

When buffer space is exhausted, victim selection m_{drop} is executed over candidate set:

$$\mathcal{B}_{\text{cand}} = \big\{ m \in \mathcal{B}_i(t) \ \big|\ m \notin \mathcal{M}_{\text{sending}} \ \wedge\ m \in \mathcal{C}_1 \big\} \tag{34}$$

according to lexicographical order:

$$m_{\text{drop}} = \arg \min_{m \in \mathcal{B}_{\text{cand}}} \big\langle C(m),\ \text{ttl}(m) \big\rangle \tag{35}$$

Theorem 3 (Inviolable Zero-Drop Telemetry Theorem)**.** *Under arbitrary traffic arrival rates $\lambda_i(t)$, network contention, and buffer saturation:*

$$\mathbb{P}\big(\mathrm{Drop}(m) \;\big|\; m \in \mathcal{C}_0\big) \equiv 0, \quad \forall t \geq 0 \tag{36}$$

Proof. By algorithmic construction in `getNextMessageToRemove()`:

$$\forall m \in \mathcal{B}_i(t), \quad m \in \mathcal{C}_0 \implies m \notin \mathcal{B}_{\mathrm{cand}}$$

Since the victim selection operator $\arg\min$ is strictly restricted to domain $\mathcal{B}_{\mathrm{cand}}$, the probability that an element of $\mathcal{C}_0$ is selected for eviction is identically zero. □

8 Closed-Loop Dynamic Stability Analysis

8.1 Continuous Fluid Queue Dynamics

The continuous buffer state $S_i(t) = S_{\max,i} \cdot Q_i(t)$ obeys the differential equation:

$$\frac{dS_i(t)}{dt} = \lambda_{i,\mathrm{gen}}(t) + \sum_{j \in \mathcal{A}_i(t)} \lambda_{ji,\mathrm{in}}\big(\mathbf{u}_i(t)\big) - \sum_{j \in \mathcal{A}_i(t)} \mu_{ij,\mathrm{out}}\big(\mathbf{u}_i(t)\big) - \delta_i(t) \tag{37}$$

where $\delta_i(t)$ is the voluntary shedding rate of bulk science replicas.

8.2 Lyapunov Stability Proof

Theorem 4 (Asymptotic System Stability)**.** *The closed-loop DTNMA feedback control system is asymptotically stable around nominal buffer equilibrium $Q^* = 0.70$.*

Proof. Define the quadratic Lyapunov candidate function:

$$V\big(Q_i(t)\big) = \frac{1}{2}\big(Q_i(t) - Q^*\big)^2 \tag{38}$$

The continuous time derivative is:

$$\dot{V}\big(Q_i(t)\big) = \big(Q_i(t) - Q^*\big) \cdot \dot{Q}_i(t) \tag{39}$$

Assume a severe congestion perturbation where $Q_i(t) \geq 0.92$. Then $(Q_i(t) - Q^*) \geq 0.22 > 0$. At this boundary, Priority 100 Rule fires, actuating:

1) $M_{\mathrm{ingress}} = 1 \implies \Gamma(m, i) =$ DENIED_NO_SPACE for bulk, forcing $\lambda^{\mathcal{C}_1}_{ji,\mathrm{in}} \to 0$.
2) $B_{\mathrm{backpressure}} = 1 \implies$ suppression of inbound transfers from any neighbor j with $Q_j \geq Q_i - 0.15$.
3) $L_{\mathrm{spray}} \leftarrow 2 \implies$ reduction in outbound replication burden.

Under these controls, the net derivative simplifies to:

$$\dot{Q}_i(t) = \frac{1}{S_{\max,i}} \left(\lambda^{\mathcal{C}_0}_{i,\mathrm{gen}} - \mu^{\mathcal{C}_0}_{i,\mathrm{out}} - \delta^{\mathcal{C}_1}_i\right)$$

Since bulk eviction rate $\delta^{\mathcal{C}_1}_i + \mu_{i,\mathrm{out}} \gg \lambda^{\mathcal{C}_0}_{i,\mathrm{gen}}$ during severe contention:

$$\dot{Q}_i(t) < 0 \implies \dot{V}\big(Q_i(t)\big) < 0, \quad \forall Q_i(t) \geq 0.92$$

By Lyapunov's direct method, $Q_i(t)$ is strictly dissipative and converges asymptotically into the invariant set $\Omega = \{Q_i \mid Q_i \leq 0.92\}$. □

Algorithm 1 Pre-Cached Policy-Governed Forwarding

Require: Active connections $\mathcal{A}_i(t)$, Buffer $\mathcal{B}_i(t)$
Ensure: Transmitted bundle m^* or $\emptyset$
1: $\mathcal{A}_{\mathrm{ready}} \leftarrow \{c \in \mathcal{A}_i(t) \mid c.\mathrm{isReady}()\}$
2: **if** $\mathcal{A}_{\mathrm{ready}} = \emptyset$ **or** $\mathcal{B}_i(t) = \emptyset$ **then return** $\emptyset$
3: **end if**
4: **Pre-cache Peer Metadata once:**
5: **for** $k \leftarrow 1$ to $|\mathcal{A}_{\mathrm{ready}}|$ **do**
6: $\quad \mathrm{peerRouters}[k] \leftarrow \mathrm{getRouter}(\mathcal{A}_{\mathrm{ready}}[k])$
7: $\quad Q_{\mathrm{peer}}[k] \leftarrow \mathrm{peerRouters}[k].\mathrm{getBufferOccupancyRatio}()$
8: **end for**
9: $\mathcal{B}_{\mathrm{sorted}} \leftarrow \mathrm{sortDualTier}(\mathcal{B}_i(t))$ ▷ $\mathcal{O}(|\mathcal{B}| \log |\mathcal{B}|)$
10: **for** $m \in \mathcal{B}_{\mathrm{sorted}}$ **do**
11: $\quad$ **if** $\mathrm{isSending}(m.\mathrm{id})$ **then continue**
12: $\quad$ **end if**
13: $\quad$ **for** $k \leftarrow 1$ to $|\mathcal{A}_{\mathrm{ready}}|$ **do**
14: $\quad\quad j \leftarrow \mathrm{peerHosts}[k]$
15: $\quad\quad$ **if** $d(m) = j$ **then** ▷ Unconditional Direct Delivery
16: $\quad\quad\quad$ **return** $\mathrm{startTransfer}(m, \mathcal{A}_{\mathrm{ready}}[k])$
17: $\quad\quad$ **end if**
18: $\quad\quad$ **if** $C(m) > 1$ **then** ▷ Spray Phase
19: $\quad\quad\quad$ **if** $m \in \mathcal{C}_1 \wedge B_{\mathrm{bp}} \wedge Q_{\mathrm{peer}}[k] \geq Q_i + 0.15$ **then continue**
20: $\quad\quad\quad$ **end if**
21: $\quad\quad\quad$ **return** $\mathrm{startTransfer}(m, \mathcal{A}_{\mathrm{ready}}[k])$
22: $\quad\quad$ **else** ▷ Cognitive Focus Phase
23: $\quad\quad\quad$ **if** $\Psi_{\mathrm{focus}}(m, i, j) = \mathrm{TRUE}$ **then**
24: $\quad\quad\quad\quad$ **return** $\mathrm{startTransfer}(m, \mathcal{A}_{\mathrm{ready}}[k])$
25: $\quad\quad\quad$ **end if**
26: $\quad\quad$ **end if**
27: $\quad$ **end for**
28: **end for**
29: **return** $\emptyset$

9 Algorithmic Implementation and Complexity

Standard DTN routing engines evaluate peer buffers repeatedly within nested loops over all messages and all connections, incurring computational complexity of $\mathcal{O}(|\mathcal{B}_i| \cdot |\mathcal{A}_i|)$.

As formalized in Algorithm 1, DTNMA introduces a vector pre-caching stage (Lines 4–7), reducing peer inspection complexity to $\mathcal{O}(|\mathcal{A}_i|)$. The overall execution cost per update cycle is:

$$\mathcal{T}_{\mathrm{cycle}} = \mathcal{O}(|\mathcal{A}_i|) + \mathcal{O}(|\mathcal{B}_i| \log |\mathcal{B}_i|) + \mathcal{O}(|\mathcal{B}_i| \cdot |\mathcal{A}_i|)_{\mathrm{eval}} \tag{40}$$

where the evaluation term consists solely of $\mathcal{O}(1)$ primitive array reads without virtual method overhead.

10 Summary Mapping to Open-Source Implementation

Table 1 provides direct references between the mathematical derivations established in this work and the corresponding Java source implementations in the autonomic DTNMA routing engine.

TABLE 1
Mathematical Formulation to Codebase Implementation Mapping

Mathematical Formulation	Operational Semantics	Source Implementation Reference
$Q_i(t) = \frac{S_{\max,i} - S_{\text{free},i}(t)}{S_{\max,i}}$	Buffer occupancy ratio	`DtnmaRouter.java` (`getBufferOccupancyRatio`)
$\dot{Q}_i(t) = \frac{Q_i(t) - Q_i(t-\Delta t)}{\Delta t}$	Finite-difference queue drift	`EddTelemetrySampler.java` (`sampleNow`)
$v_i(t) = \frac{\|\mathbf{p}_i(t) - \mathbf{p}_i(t-\Delta t)\|_2}{\Delta t}$	Kinematic Euclidean speed	`EddTelemetrySampler.java` (`sampleNow`)
$D_i(t) = \|\mathcal{H}_i(t, W) \cup \mathcal{A}_i(t)\|$	Sliding-window contact density ($W = 300\,\text{s}$)	`EddTelemetrySampler.java` (`getNodeDensity`)
$P_{(i,j)} \leftarrow P + (1-P)P_{\text{init}}$	PRoPHET direct encounter update	`EddTelemetrySampler.java` (`updateDeliveryPredFor`)
$P_{(i,k)} \leftarrow P + (1-P)P_{(i,j)}P_{(j,k)}\beta$	Transitive delivery predictability	`EddTelemetrySampler.java` (`updateTransitivePreds`)
$k^* = \arg\max \pi(R_k)$ s.t. $\Phi_k = \text{TRUE}$	Priority rule arbitration	`DtnmaRouter.java` (`evaluateAutonomyEngine`)
$C_{\text{sender}} = \lceil C/2 \rceil,\ C_{\text{recv}} = \lfloor C/2 \rfloor$	Binary spray quota splitting	`DtnmaRouter.java` (`transferDone`, `messageTransferred`)
$\Psi_{\text{focus}}(m, i, j)$	Cognitive handoff gating ($C = 1$)	`DtnmaRouter.java` (`tryPolicyGovernedForwarding`)
$\mathcal{B}_i \leftarrow \mathcal{B}_i \setminus \{m\}$	Single-copy custodial relinquishment	`DtnmaRouter.java` (`transferDone`)
$G(m, i, j) \in \{0, 1\}$	Backpressure gradient gating	`DtnmaRouter.java` (`tryPolicyGovernedForwarding`)
$m_1 \prec m_2$	Dual-tier priority HoL sorting	`DtnmaRouter.java` (`sortMessagesDualTier`)
$\Gamma(m, i) \in \{\text{OK}, \text{DENIED}\}$	Constructive ingress admission control	`DtnmaRouter.java` (`checkReceiving`)
$m \in \mathcal{C}_0 \implies m \notin \mathcal{B}_{\text{cand}}$	Inviolable zero-drop storage exclusion	`DtnmaRouter.java` (`getNextMessageToRemove`)

11 Conclusion

In this paper, we formulated the complete mathematical foundation of the autonomic DTNMA routing framework. By formalizing strongly-typed EDD telemetry sampling, continuous encounter predictability recurrence, binary token splitting dynamics, and constructive ingress flow control, the system achieves self-governing adaptation across heterogeneous operational regimes. We proved the Token Conservation Invariance, the Single-Copy Invariant, the Inviolable Zero-Drop Theorem for mission-critical telemetry, and demonstrated asymptotic Lyapunov stability under severe buffer congestion.

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